

ITY

L.N. Mithila University, Darbhanga

A. N. D. College

PROJECT REPORT
ON

AI ETHICS & RESPONSIBLE TECH POLICY RESEARCH

*Human Rights, Fairness, Transparency, Accountability, Privacy, Safety,
Human Oversight, AI Risk Governance, Regulation and Policy Impact Assessment*

Submitted in partial fulfilment of the requirements for the award of the degree of

B.A. (General) (Hons.)

Internship Domain: AI Ethics & Responsible Tech Policy Research

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DECLARATION

I, Raunak testing, Registration No. 4565665, Roll No. 4565665, a student of B.A. (Course) in AI Ethics & Responsible Tech Policy Research (Internship Domain) at A. N. D. College (College), affiliated to L.N. Mithila University, Darbhanga (University), hereby declare that this project

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1. The report has not been submitted elsewhere for the award of any degree or diploma.
 2. Policy and regulatory claims are based on cited official or institutional sources and are stated with their time and scope limitations.
 3. The case study, risk scores and organisational findings are fictional and are used only to demonstrate responsible-AI policy research methods.
 4. No political party, candidate or electoral choice is endorsed or opposed; the report compares documented policy frameworks descriptively.
 5. The report distinguishes ethical principles, voluntary standards and binding legal requirements rather than treating them as equivalent.

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ABSTRACT

Artificial intelligence increasingly influences decisions, information flows, work, education, public services and commercial activity. AI ethics asks what values should guide these systems; responsible technology governance converts those values into organisational controls, standards, rights, accountability processes and law. The core challenge is not only whether an AI model is technically accurate, but whether the overall socio-technical system is fair, safe, privacy-respecting, explainable enough for its context, contestable, secure and subject to meaningful human responsibility.

This project report examines AI ethics and responsible-tech policy research through international and Indian frameworks. It covers human rights and dignity, fairness and non-discrimination, transparency and explainability, accountability, privacy and data governance, safety and robustness, human oversight, information integrity, accessibility, labour effects and environmental sustainability. It compares UNESCO's 2021 Recommendation on the Ethics of Artificial Intelligence, the OECD AI Principles updated in 2024, NIST's AI Risk Management Framework and Generative AI Profile, the European Union AI Act, and India's responsible-AI and data-protection policy landscape.

A hypothetical case study applies an AI Impact Assessment to "CareerBridge", an AI-assisted internship recommendation platform for university students. The assessment identifies risks relating to proxy discrimination, opaque ranking, sensitive data, feedback loops, accessibility, vendor dependence and over-reliance on automation. Mitigations include purpose limitation, human review, outcome testing, data minimisation, documentation, appeal channels, security controls and periodic monitoring. The report concludes that responsible AI is best understood as a lifecycle governance discipline combining policy, engineering, legal compliance, organisational accountability and stakeholder participation.

Keywords: AI ethics, responsible AI, trustworthy AI, AI governance, fairness, transparency, accountability, privacy, human oversight, AI risk management, EU AI Act, UNESCO AI ethics, OECD AI Principles, NIST AI RMF, India responsible AI, DPDP Act.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
AIA	AI Impact Assessment
DPDP	Digital Personal Data Protection
GPAI	General-Purpose AI
NIST	National Institute of Standards and Technology
OECD	Organisation for Economic Co-operation and Development
RAI	Responsible Artificial Intelligence
RMF	Risk Management Framework
TEVV	Testing, Evaluation, Verification and Validation
UNESCO	United Nations Educational, Scientific and Cultural Organization

CHAPTER 1

Introduction

1.1 Background

Artificial intelligence systems can support diagnosis, education, accessibility, scientific discovery, productivity and public administration. The same systems can also create or amplify harms when data are unrepresentative, objectives are poorly defined, models are insecure, outputs are over-trusted, or decision processes lack accountability. Responsible AI therefore examines the full system: people, institutions, data, models, interfaces, incentives, vendors, laws and affected communities.

RESPONSIBLE AI & TECH POLICY ECOSYSTEM

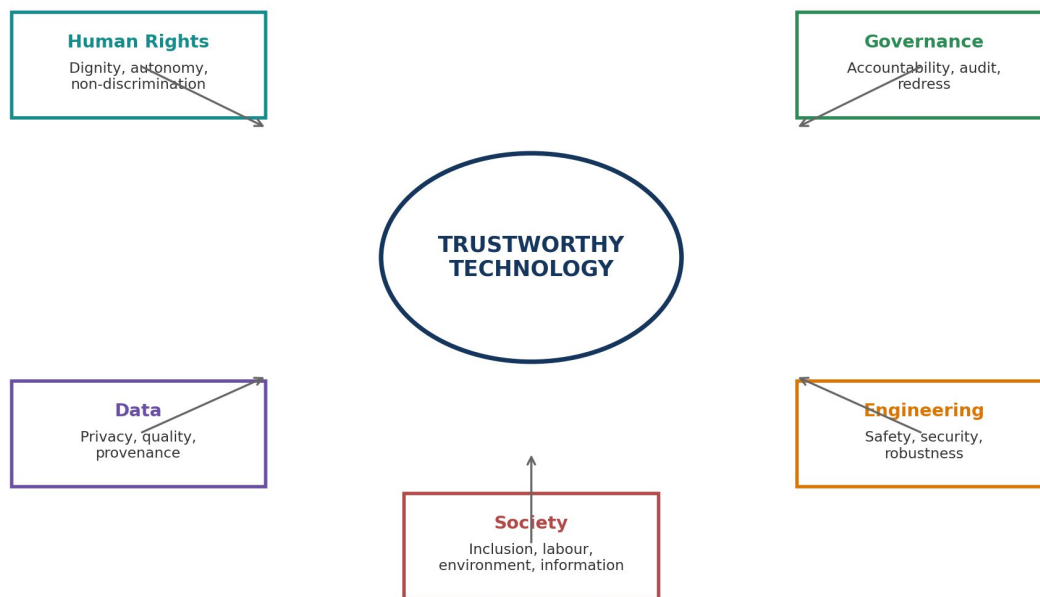


Figure 1: Responsible AI and tech-policy ecosystem

1.2 AI Ethics and Responsible Technology

AI ethics concerns principles and values that should guide AI design, deployment and use. Responsible technology policy converts those values into operational requirements such as documentation, testing, human review, data governance, access controls, audit, incident response, complaint handling and regulatory compliance. Ethics asks what ought to be protected; governance asks who must do what, when, with what evidence and with what remedy if harm occurs.

1.3 Why Policy Research Matters

AI governance is evolving quickly and uses different instruments: international recommendations, voluntary standards, technical frameworks, national laws, sector rules, procurement conditions and organisational policies. A policy researcher must distinguish their legal status and jurisdiction. An ethics principle may influence policy without creating a directly enforceable legal duty; a binding law may apply only to specific actors, uses or territories.

1.4 Objectives of the Project

1. To explain core principles of AI ethics and responsible technology.
2. To classify AI risks across rights, fairness, privacy, safety, information integrity, labour and the environment.
3. To compare major international governance frameworks without treating them as legally equivalent.
4. To study India's responsible-AI and digital data-protection policy landscape.
5. To develop a practical AI Impact Assessment and organisational control framework.
6. To apply the framework to a hypothetical AI-assisted internship recommendation system.
7. To identify research limitations and future directions in responsible AI policy.

1.5 Scope and Limitations

This is an academic policy-research report, not legal advice. The regulatory landscape can change after publication. Jurisdiction-specific compliance questions require checking the latest statute, rules, regulator guidance and sector-specific requirements. The case study is fictional and does not evaluate a real company or system.

1.6 Organisation of the Report

Chapter 2 covers ethical foundations and risks. Chapter 3 compares international frameworks. Chapter 4 focuses on India. Chapter 5 explains policy research and impact assessment. Chapter 6 presents the case study. Chapter 7 compares results and governance functions, and Chapter 8 concludes.

CHAPTER 2

Ethical Foundations and AI Risk Taxonomy

2.1 Human Rights and Human Dignity

A human-rights approach asks whether AI affects equality, privacy, freedom of expression, access to services, due process, autonomy or other protected interests. UNESCO's 2021 Recommendation places human rights and dignity at the core of AI ethics. Rights-based analysis is especially important when AI influences high-impact decisions or affects vulnerable groups.

2.2 Fairness and Non-Discrimination

AI can create unequal outcomes through historical bias, sampling imbalance, proxy variables, label quality, accessibility barriers or deployment choices. Fairness is not one universal mathematical metric. Policy analysis should identify the affected population, decision context, relevant legal norms and trade-offs before choosing tests. A system can satisfy one statistical fairness measure and fail another.

2.3 Transparency and Explainability

Transparency can include disclosure that AI is being used, information about purpose and limitations, documentation for auditors, and explanations for people affected by a decision. Explainability should be useful to its audience. A technical feature-attribution plot may help a developer but may not satisfy a person seeking to understand why an application was rejected.

2.4 Accountability

Accountability requires identifiable owners, documented decisions, audit trails, escalation routes and remedies. Responsibility should not disappear merely because a model or third-party vendor produced a recommendation. Organisations need defined decision rights across product, engineering, legal, security, compliance and business teams.

2.5 Privacy and Data Governance

Responsible AI data governance includes purpose limitation, data minimisation, access control, retention limits, provenance, quality, security and lawful processing. AI introduces additional concerns such as training-data reuse, model memorisation, inference of sensitive attributes and data leakage through prompts or outputs.

2.6 Safety, Security and Robustness

Safety addresses foreseeable harm from intended and unintended behaviour. Security addresses malicious exploitation, including prompt injection, data poisoning, model theft and unsafe tool access. Robustness considers performance under realistic changes or failures. NIST's AI RMF treats trustworthy AI as a risk-management problem across design, development, deployment and use.

2.7 Human Oversight and Contestability

Human oversight is meaningful only if reviewers have authority, time, information and competence to intervene. Rubber-stamping an automated recommendation is not strong oversight. Contestability requires a path for affected people to question, correct or appeal consequential outcomes where appropriate.

CORE RESPONSIBLE AI PRINCIPLES



Figure 3: Core responsible AI principles

2.8 Information Integrity and Generative AI

Generative AI introduces risks including fabricated content, impersonation, manipulation, low-quality synthetic information and automated amplification. The OECD's 2024 revision explicitly strengthened attention to information integrity and generative-AI-related risks. Governance may include provenance practices, disclosure, moderation, abuse monitoring and human verification of factual content.

2.9 Labour and Human Capability

AI can automate tasks, augment workers, change skill requirements and alter monitoring or performance-management practices. Ethical analysis considers job quality, surveillance, reskilling, worker participation and distribution of benefits. Effects differ by occupation and deployment model; broad claims about universal job loss or universal productivity gains should be treated cautiously.

2.10 Environmental Sustainability

AI systems use energy, hardware, data-centre infrastructure and water. Environmental analysis can include model-training and inference energy, hardware lifecycle, efficiency, reuse and whether the AI application produces environmental benefits that exceed its costs. The OECD's updated AI Principles explicitly reference environmental sustainability.

2.11 AI Harm Taxonomy

Risk Area	Examples	Evidence / Control
Fairness	Disparate error rates; proxy discrimination	Segmented evaluation; representative data;

		outcome monitoring
Privacy	Overcollection; unauthorised reuse; prompt leakage	Minimisation; access control; retention; secure processing
Safety	Unsafe recommendation; automation error	Hazard analysis; fallback; human review; testing
Security	Prompt injection; data poisoning; model theft	Threat modelling; isolation; red teaming; monitoring
Transparency	Hidden AI use; inadequate explanation	Notice; documentation; decision explanation
Accountability	No clear owner or appeal route	RACI; audit logs; grievance / escalation
Information integrity	Hallucination; synthetic deception	Verification; provenance; abuse detection
Social / labour	Worker surveillance; exclusion	Impact assessment; consultation; proportionality
Environment	High compute / energy footprint	Efficiency measurement; model selection; infrastructure controls

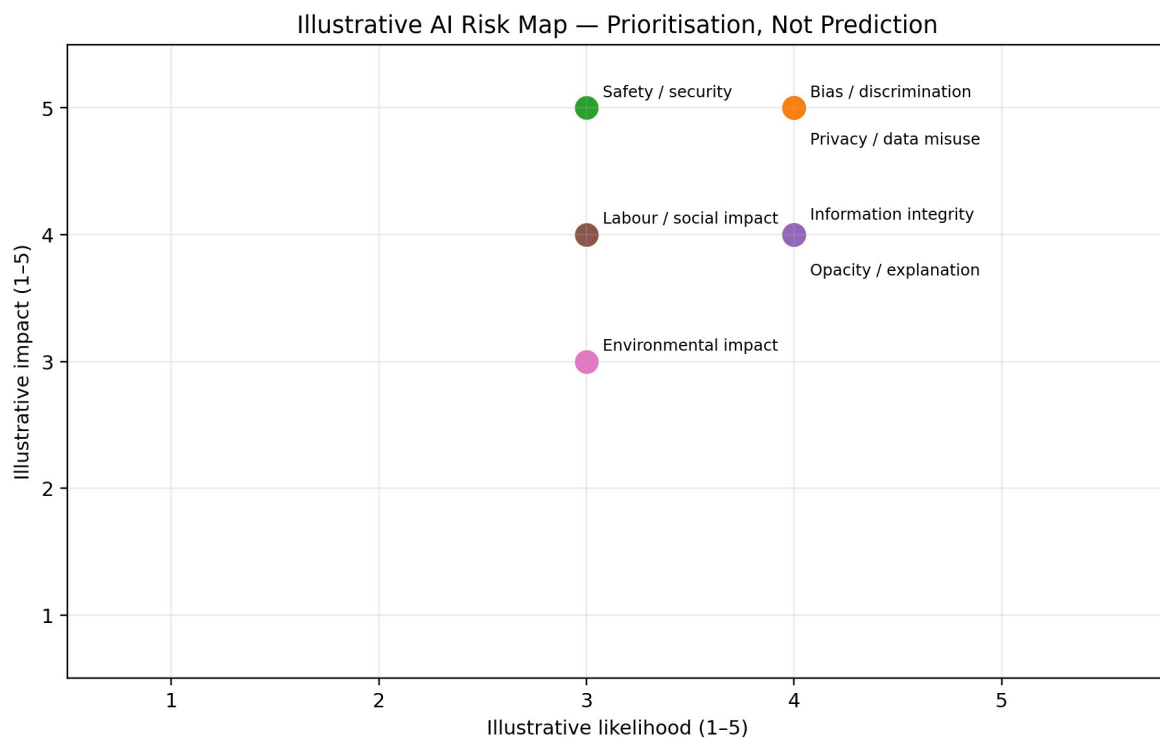


Figure 4: Illustrative AI risk map — prioritisation, not prediction

CHAPTER 3

International AI Governance Frameworks

3.1 UNESCO Recommendation on the Ethics of AI

UNESCO's Recommendation on the Ethics of Artificial Intelligence was adopted on 23 November 2021 by UNESCO's 193 Member States. It is a global normative instrument rather than a directly self-executing national statute. It centres human rights and dignity and includes policy action areas such as ethical impact assessment, data policy, environment, education, labour, health, culture and communication.

Policy research note: Because UNESCO's Recommendation is an international normative instrument, implementation depends on states and institutions translating it into laws, policies, programmes and organisational practice.

3.2 OECD AI Principles

The OECD AI Principles were first adopted in 2019 and updated in May 2024. The updated principles retain five values-based principles— inclusive growth and well-being; respect for rule of law, human rights and democratic values including fairness and privacy; transparency and explainability; robustness, security and safety; and accountability—along with recommendations to governments. The 2024 revision responds to general-purpose and generative AI, information integrity, privacy, intellectual property, safety and environmental sustainability.

3.3 NIST AI Risk Management Framework

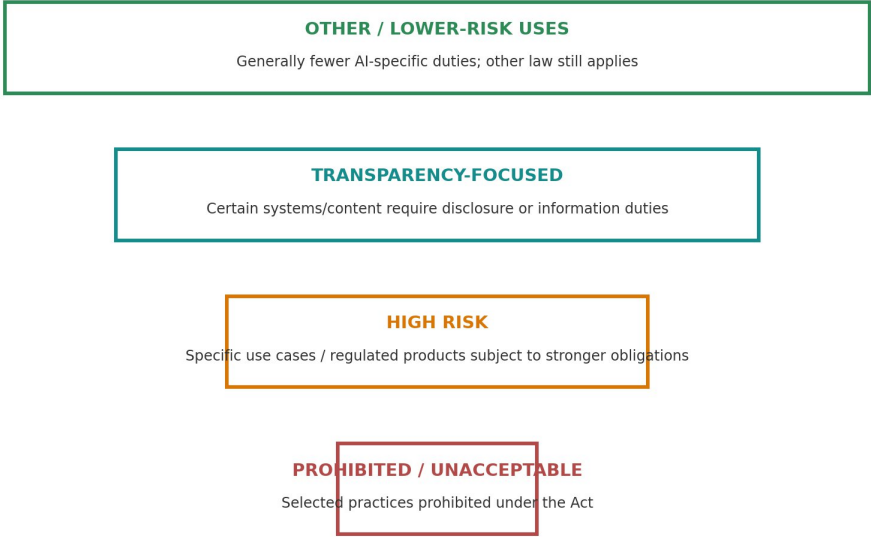
NIST released AI RMF 1.0 in January 2023 as a voluntary framework to help organisations manage AI risks to individuals, organisations and society. The framework is commonly organised around the functions GOVERN, MAP, MEASURE and MANAGE. NIST released a Generative AI Profile (NIST AI 600-1) in July 2024. As of September 2026, NIST states that AI RMF 1.0 is being revised.

Policy research note: NIST AI RMF is a voluntary risk-management resource, not a general AI law. Organisations may use it alongside applicable legal and sector obligations.

3.4 European Union AI Act

The EU AI Act entered into force on 1 August 2024 and follows a risk-based regulatory model. The Commission states that prohibited-practice and AI-literacy provisions began applying on 2 February 2025, governance and general-purpose AI obligations on 2 August 2025, and the Act became generally applicable on 2 August 2026 subject to exceptions and transition rules. Current Commission guidance also reflects later adjustments to the timing of certain high-risk provisions.

EU AI ACT — RISK-BASED STRUCTURE (SIMPLIFIED)



The Act also contains separate obligations for general-purpose AI models and governance/enforcement mechanisms.

Figure 6: EU AI Act risk-based structure (simplified)

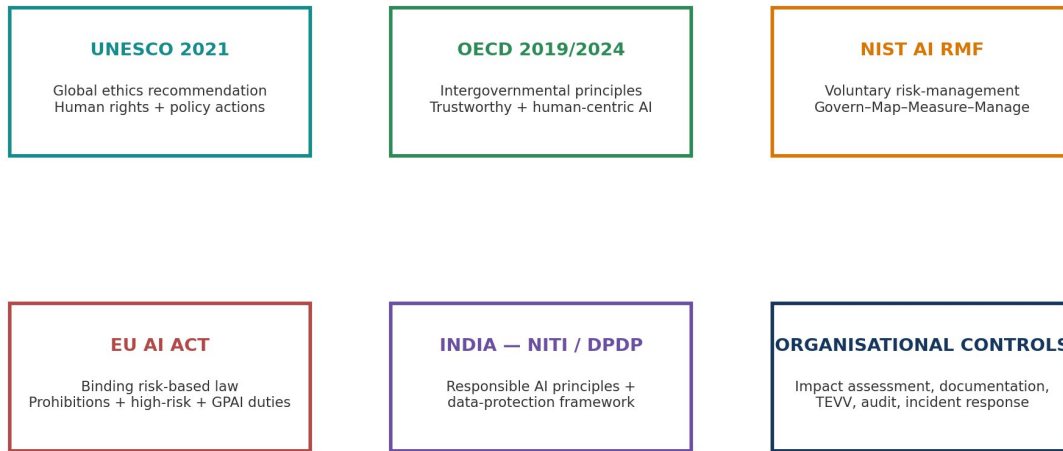
Selected Milestone	Date / Status as stated by European Commission
AI Act enters into force	1 August 2024
Definitions, AI literacy and selected prohibitions apply	2 February 2025
Governance and GPAI obligations apply	2 August 2025
General applicability	2 August 2026, subject to exceptions
Certain high-risk rules	Later transition dates apply; current Commission guidance should be checked for the specific category

3.5 General-Purpose AI Models under the EU AI Act

The Commission states that providers of general-purpose AI models face obligations including technical documentation, a copyright policy and a public summary of training content. Providers of GPAI models with systemic risk face additional duties including risk assessment and mitigation, incident reporting and cybersecurity protections.

3.6 Framework Comparison

SELECTED RESPONSIBLE AI GOVERNANCE FRAMEWORKS



Frameworks differ in legal status, scope and enforcement; they should not be treated as interchangeable.

Figure 5: Selected responsible AI governance frameworks

Framework	Type / Legal Status	Primary Focus	Operational Value
UNESCO Recommendation	International normative recommendation	Human rights, ethics and policy action	Ethical impact and policy design
OECD AI Principles	Intergovernmental recommendation / principles	Trustworthy human-centric AI	Policy interoperability and common principles
NIST AI RMF	Voluntary technical / organisational framework	AI risk management	Lifecycle governance and risk controls
EU AI Act	Binding EU regulation	Risk-based legal obligations	Prohibitions, high-risk duties, transparency, GPAI
Organisation policy	Internal governance instrument	Context-specific controls	Approval, documentation, testing, monitoring and accountability

3.7 Interoperability and Limits

The frameworks overlap on fairness, transparency, safety and accountability, but their scope and legal effects differ. A multinational organisation may use a common internal risk framework while mapping each deployment to local legal obligations. Interoperability does not mean that compliance with one voluntary framework guarantees compliance with another jurisdiction's law.

CHAPTER 4

India — Responsible AI and Data Governance

4.1 NITI Aayog Responsible AI for All

NITI Aayog published “Approach Document for India Part 1 — Principles for Responsible AI” in February 2021 and Part 2 on operationalising responsible-AI principles in August 2021. The work sought to ground responsible AI in the Indian legal and social context while balancing innovation and risk governance. It is a policy approach document, not a standalone AI statute.

4.2 Principles in the Indian Approach

The NITI Aayog work addresses themes including safety and reliability, equality, inclusivity and non-discrimination, privacy and security, transparency, accountability and protection/reinforcement of positive human values. Later use-case work, including on facial recognition, examined how such principles translate into specific design and deployment risks.

4.3 Digital Personal Data Protection Act, 2023

India’s Digital Personal Data Protection Act, 2023 provides a legal framework for processing digital personal data. AI systems that process personal data may need to be assessed under the DPDP framework in addition to other applicable laws and sector rules. The Act is not an AI-specific law, so an AI ethics review should not be reduced to data-protection compliance alone.

4.4 Digital Personal Data Protection Rules, 2025

MeitY notified the Digital Personal Data Protection Rules, 2025 in November 2025 with phased commencement. The notification states that Rules 1, 2 and 17–21 commenced on publication; Rule 4 is to commence one year after publication; and Rules 3, 5–16, 22 and 23 are to commence eighteen months after publication. This phased structure is relevant to organisations planning data-governance compliance as of September 2026.

Policy research note: Because the DPDP Rules have staged commencement, a policy report should state which provisions are currently effective rather than describing the entire ruleset as if every duty already applied on the same date.

4.5 IndiaAI Mission and Safe & Trusted AI

The IndiaAI Mission includes a “Safe & Trusted AI” pillar. By 2026, MeitY/IndiaAI materials show work toward an IndiaAI Safety Institute and responsible-AI tools and evaluation capacity. Official recruitment material for the Institute describes responsibilities including model evaluation, red-teaming, risk assessment, AI-safety guidelines and standards, and research contextualised to India’s social, economic, cultural and linguistic needs.

INDIA — RESPONSIBLE AI / DATA GOVERNANCE STACK (SIMPLIFIED)

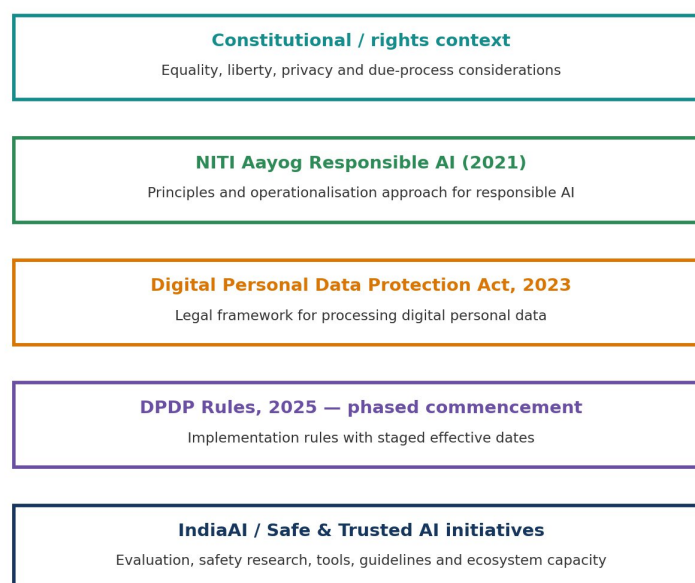


Figure 7: India responsible AI / data governance stack (simplified)

4.6 Policy Implications for Organisations in India

- Distinguish general AI ethics principles from legally binding data, consumer, cyber, sector and contractual obligations.
- Map personal-data processing to the DPDP Act and the phased DPDP Rules as they come into force.
- Use impact assessment for sensitive or high-impact AI even where no single horizontal AI law applies.
- Maintain human accountability for consequential decisions and provide escalation / correction channels where appropriate.
- Assess Indian language, disability, regional and socio-economic inclusion rather than relying only on global benchmark averages.
- Monitor evolving IndiaAI safety, evaluation and governance guidance.

4.7 India Policy Source Table

Source	Status / Role	Relevance
NITI Aayog Responsible AI Part 1 (2021)	Policy principles	Ethical principles for responsible AI in India
NITI Aayog Responsible AI Part 2 (2021)	Operationalisation approach	Institutional / enforcement mechanisms
DPDP Act, 2023	Binding law	Digital personal-data processing
DPDP Rules, 2025	Subordinate legislation with phased commencement	Operational data-protection requirements
IndiaAI Safe & Trusted AI / Safety Institute initiatives	Government programme / institutional capacity	Evaluation, safety research, tools, guidelines and standards

CHAPTER 5

Responsible AI Policy Research Methodology

5.1 Research Design

This report uses descriptive policy analysis, comparative framework analysis and an applied AI Impact Assessment. Primary preference is given to official legal texts, regulator or government guidance, intergovernmental standards and technical standards bodies. Secondary commentary is useful for interpretation but should not replace primary sources when verifying legal status or dates.

PROJECT REPORT METHODOLOGY

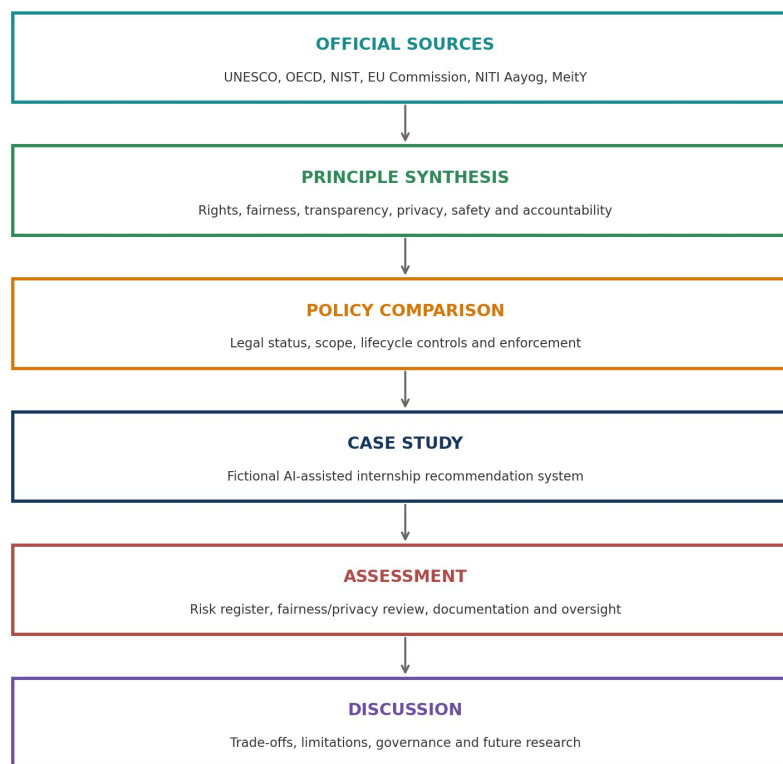


Figure 9: Methodology adopted for the project

5.2 Evidence Hierarchy

Evidence Level	Examples	Use
1. Binding primary law	Statute, regulation, official gazette	Verify legal duties, scope and dates
2. Official implementation guidance	Regulator / ministry / Commission guidance	Interpret implementation and timelines
3. Intergovernmental recommendations	UNESCO, OECD	Principles and policy design
4. Technical frameworks / standards	NIST, ISO/IEC, sector standards	Risk-management and engineering practice
5. Research and civil-society evidence	Peer-reviewed studies, audits, impact reports	Empirical effects and critiques
6. Vendor / media material	Product docs, news, blogs	Context; verify important claims

independently

5.3 AI Impact Assessment

An AI Impact Assessment documents the intended purpose, affected groups, decision context, data, model, expected benefits, risks, controls and residual risk. It can be conducted before procurement/deployment and repeated after significant changes. It is not a guarantee of safety; its value depends on evidence quality, independent challenge and follow-through.

AI IMPACT ASSESSMENT WORKFLOW



Figure 8: AI Impact Assessment workflow

5.4 System Boundary

The assessment should include more than the model. Inputs, data pipelines, prompts, retrieval systems, human reviewers, user interface, thresholds, vendor services, downstream actions and appeal mechanisms can all affect outcomes. Many real harms arise from deployment design even when model benchmark performance appears acceptable.

5.5 Stakeholder Analysis

- Direct users and people subject to AI-supported decisions.
- Groups that may be indirectly affected or underrepresented in data.
- Employees operating or reviewing the system.
- Security, legal, privacy, accessibility and compliance teams.
- Vendors and upstream model/data providers.
- Regulators, auditors and public-interest stakeholders where relevant.

5.6 Risk Rating

Risk rating should combine severity, likelihood, detectability and exposure, but numeric scores should not create false precision. Some rights impacts require special treatment even when likelihood is uncertain. A risk register should record evidence, control owner, due date, residual risk and decision status.

5.7 Fairness Evaluation

Fairness evaluation begins with the decision and affected population. It can include performance by subgroup, selection/recommendation rates, error patterns, accessibility and qualitative review. Sensitive attributes may be necessary for auditing but themselves require careful governance. The goal is to identify unjustified disparities, not to mechanically equalise every metric.

5.8 Transparency and Documentation

- Purpose and prohibited uses.
- Data sources, limitations and retention.
- Model / system version and vendor dependencies.
- Evaluation methods and known limitations.
- Human oversight and escalation paths.
- User disclosures and explanation process.
- Incident, complaint and change logs.

5.9 Testing, Evaluation, Verification and Validation

TEVV should reflect the deployment context. Useful testing can include functional accuracy, subgroup performance, adversarial testing, prompt-injection testing, privacy leakage, hallucination / factuality checks, accessibility, human-factors evaluation, load/failure modes and post-deployment monitoring. NIST's AI RMF and related resources provide a structured vocabulary for this work.

5.10 Governance Decision

An impact assessment can support several outcomes: approve, approve with conditions, redesign, limit to decision support, pilot with monitoring, postpone, or stop. Policy research should document the reasons and evidence rather than presenting a simple ethics checklist as automatic approval.

CHAPTER 6

Case Study — AI-Assisted Internship Recommendation System

6.1 Fictional System Overview

“CareerBridge” is a fictional university platform that recommends internships to students. It uses education history, stated skills, preferences, location constraints and employer requirements to rank opportunities. The system does not make final hiring decisions, but it can influence which internships students see first. Because ranking affects opportunity visibility, the project treats it as a meaningful fairness and transparency problem rather than a simple entertainment recommendation.

6.2 Intended Purpose and Boundaries

- Purpose: help students discover relevant internships and reduce search effort.
- Not permitted: automatically reject a student from an employer process.
- Not permitted: infer caste, religion, health status or other sensitive traits for ranking.
- Human choice: students can browse all eligible opportunities independent of ranking.
- Appeal / feedback: students can report irrelevant or unfair recommendations and edit profile data.

6.3 Data Inventory

Data Element	Need / Risk	Control
Course / branch	Relevant to some opportunity requirements	Use only where job requirement is legitimate
Skills / certifications	Core relevance signal	Allow correction; verify where needed
Location preference	Supports feasible recommendations	Do not use precise location when broad city preference is enough
Past applications	Can improve personalisation but create feedback loops	Limit weight; avoid penalising exploration
College identity	Potential proxy for socio-economic advantage	Test impact; consider removal / constrained use
Name / photo	Not needed for ranking	Exclude from model features
Disability accommodation preference	Potentially sensitive	Separate support workflow; do not reduce ranking

6.4 Primary Risks

Risk	How It Could Occur	Initial Rating
Proxy discrimination	College / postcode correlates with socio-economic status	High
Popularity feedback loop	Already-visible internships become even more visible	High
Opaque ranking	Students cannot understand why opportunities differ	Medium-High
Sensitive-data leakage	Profile data exposed through logs / prompts	High
Accessibility exclusion	Interface or ranking ignores accessibility needs	Medium
Automation bias	Counsellors over-trust ranking order	Medium-High
Vendor dependency	External model changes without notice	Medium

6.5 Fairness Review

The team tests recommendation coverage and relevance by programme, gender where lawfully available for auditing, institution type and geographic background. The objective is not identical recommendations for all groups; opportunities legitimately differ by skill and location. The review instead looks for unexplained differences after controlling for relevant eligibility factors.

6.6 Privacy Review

The platform removes names and photographs from ranking features, reduces location to user-selected preferred regions, limits retention of application history, separates analytics identifiers and restricts staff access. Prompt logs for any generative component are configured to avoid storing free-text personal data longer than necessary.

6.7 Transparency Design

- Label that recommendations are AI-assisted.
- Show high-level reasons such as “matches Python skill + remote preference”.
- Show missing profile fields that affect recommendation quality.
- Provide “show all eligible internships” independent of ranking.
- Allow correction of profile data and reporting of an irrelevant recommendation.

6.8 Human Oversight

Career counsellors can review recommendation patterns but are instructed not to treat the rank as a measure of student quality. Employer eligibility rules are separately validated so that the platform does not encode an employer’s unlawful or unjustified preference merely because it was supplied as a filter.

6.9 Security and Abuse Testing

- Access controls for student profiles and employer data.
- Tests for prompt injection if a generative assistant is used.
- Rate limits and monitoring for scraping or account abuse.
- Vendor version tracking and change notices.
- Incident runbook for data exposure or harmful recommendation behaviour.

6.10 Mitigation Plan

Control	Owner	Evidence of Completion
Remove name/photo and restrict college-proxy use	Product + ML	Feature list and ablation/fairness test
Subgroup outcome testing	ML + Responsible AI	Evaluation report with thresholds and caveats
AI-use disclosure and reasons	UX + Legal	User-interface review
Browse-all alternative	Product	Functional test
Data minimisation / retention limits	Privacy + Engineering	Data inventory and deletion tests
Counsellor training on automation bias	Operations	Training attendance + guidance
Vendor model-change governance	Procurement + Security	Contract + version / change log
Quarterly monitoring	Responsible AI committee	Dashboard + issue register

6.11 Illustrative Findings After Mitigation

The fictional review begins with eight high-priority unresolved risks and multiple fairness, privacy, documentation and oversight gaps. After redesign and control implementation, most issues are closed or reduced, while two residual risks remain open for monitoring: changing employer eligibility patterns and model drift across academic cycles. These numbers are illustrative and do not represent a real system.

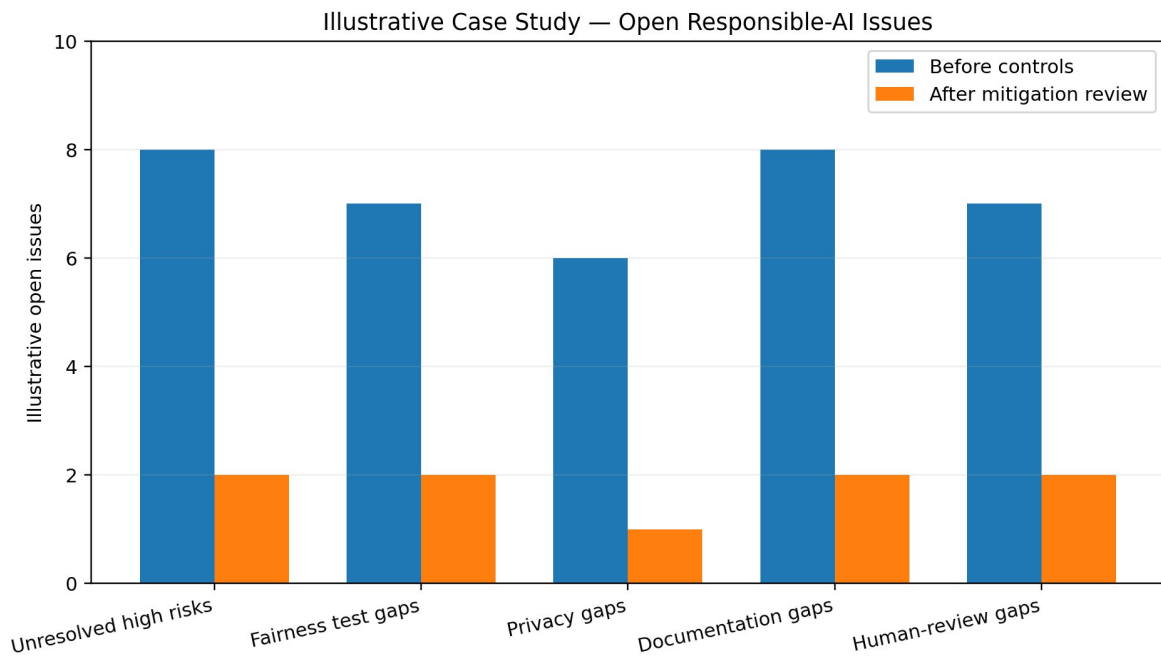


Figure 10: Illustrative case-study open issues before and after controls

6.12 Governance Decision

The fictional governance committee approves a limited pilot with conditions: no automated rejection, browse-all access, documented subgroup monitoring, data-retention controls, user correction/feedback, quarterly review and automatic re-assessment after major model or data changes. This demonstrates conditional deployment rather than a binary “ethical / unethical” label.

CHAPTER 7

Results, Comparison and Discussion

7.1 Convergence Across Frameworks

UNESCO, OECD, NIST, EU and Indian responsible-AI materials converge around several themes: human well-being, fairness, transparency, safety, accountability and responsible data use. The strongest common lesson is that trustworthiness must be operationalised across the AI lifecycle rather than added as a final ethics review.

7.2 Differences in Governance Function

Governance Function	UNESCO / OECD	NIST	EU AI Act	India Landscape
Ethical principles	Strong	Reflected in trustworthiness characteristics	Embedded in legal objectives / rights protections	NITI responsible-AI principles
Lifecycle risk process	Policy-oriented	Core strength	Mandatory controls for covered actors / uses	Developing through organisational and mission initiatives
Binding legal obligations	Not by themselves	No	Yes, within EU scope	DPDP and other applicable laws; no single equivalent horizontal AI Act cited here
Technical testing guidance	General	Strong practical framework	Standards / conformity mechanisms support implementation	Growing safe-and-trusted AI capability
Human-rights framing	Strong	Risk framing includes impacts	Fundamental-rights-oriented	Constitutional / legal context + responsible-AI principles

7.3 Principles Versus Controls

A principle such as fairness does not specify which dataset to use, what disparity threshold is acceptable, who approves exceptions or what remedy exists. Governance requires translating principles into measurable controls and decision rights. The case study illustrates this translation through feature restrictions, testing, user choice, monitoring and appeal mechanisms.

7.4 Compliance Versus Responsibility

Legal compliance is necessary where law applies, but responsible-AI governance may address risks not yet fully covered by specific law. Conversely, an internal ethics review cannot replace legal compliance. The two functions should interact: legal teams identify binding obligations, while responsible-AI governance evaluates broader lifecycle impacts and residual risks.

7.5 Risk-Based Governance

Risk-based governance allocates more scrutiny to systems with greater potential impact. This can make oversight proportionate, but it also creates classification challenges. Organisations should avoid deliberately narrowing system descriptions merely to place a use case into a lower-risk category. The system boundary and actual downstream effect matter.

7.6 The Role of Human Oversight

Human oversight is valuable when the human has context and authority. For large-scale repetitive decisions, simply inserting a human click may create automation bias rather than accountability. Better controls may include threshold design, exception routing, random audit, decision explanation and appeal.

7.7 Model and Vendor Governance

Many organisations deploy third-party foundation models or APIs. Responsible procurement therefore includes vendor documentation, security, data-use terms, model-update processes, incident notification, audit rights where feasible and exit plans. Outsourcing the model does not outsource the impact on users.

7.8 Generative AI-Specific Risks

- Confident fabrication and factual errors.
- Prompt injection and unsafe tool use.
- Leakage of confidential data through prompts, logs or outputs.
- Copyright and training-content questions.
- Impersonation, synthetic media and information-integrity risks.
- Over-reliance on generated advice in high-impact contexts.

7.9 Organisational Responsible-AI Controls

Control Area	Practical Mechanism
Governance	AI inventory, risk tiers, approval committee, named accountable owner
Data	Provenance, minimisation, quality checks, retention and access controls
Model / system	Evaluation, red teaming, robustness and security testing
Human factors	Training, override, escalation and automation-bias controls
Transparency	User notice, explanations, limitations and documentation
Monitoring	Drift, subgroup outcomes, complaints, incidents and version changes
Remedy	Correction, appeal, incident response and compensating controls
Third parties	Due diligence, contractual requirements and change governance

7.10 Research Limitations

This report compares a limited selection of influential frameworks. It does not cover every national AI law, ISO/IEC standard, sector rule or court decision. Regulatory status and implementation timelines can change. The case study is illustrative and not empirical evidence of effectiveness.

CHAPTER 8

Conclusion and Future Scope

8.1 Conclusion

- AI ethics provides values; responsible-tech governance converts them into roles, controls, evidence and remedies.
- Fairness, privacy, safety, transparency and accountability must be assessed in context rather than treated as one-time checkboxes.
- UNESCO, OECD, NIST and the EU AI Act serve different governance functions and legal statuses.
- India's responsible-AI landscape includes NITI Aayog principles, the DPDP legal framework and growing Safe & Trusted AI institutional capacity.
- AI Impact Assessments help connect system purpose, affected groups, evidence, risks and mitigation decisions.
- Human oversight must be meaningful, not symbolic.
- Third-party models require vendor and lifecycle governance.
- Responsible AI is a continuous process because data, models, users and social context change over time.

8.2 Future Scope

- AI assurance and independent audit methods with clearer evidence standards.
- Model evaluation for Indian languages and culturally diverse populations.
- Privacy-preserving evaluation and secure use of sensitive audit attributes.
- Better governance for agentic AI, tool-using systems and automated workflows.
- Content provenance and authenticity infrastructure for synthetic media.
- AI safety evaluation for critical infrastructure and public-sector deployment.
- Environmental measurement for model training, inference and hardware lifecycle.
- Participatory policy design involving affected communities, workers and civil society.

8.3 Final Note

Responsible AI policy research is most useful when it avoids both extremes: treating AI as uniquely beyond existing governance, or assuming that existing law and ordinary software practice automatically address every AI-specific risk. The practical task is to identify which harms are genuinely new, which are amplified by scale or opacity, which existing protections apply, and what additional controls are proportionate to the context.

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APPENDIX A: AI IMPACT ASSESSMENT TEMPLATE

Section	Questions / Evidence
System purpose	What problem is solved? What decisions/actions does AI influence?
Affected people	Who uses the system? Who is subject to it? Who may be indirectly affected?
Data	Sources, personal/sensitive data, provenance, quality, retention and access
Model / vendor	Model/version, provider, limitations, update/change process
Potential benefits	Evidence-based benefits and who receives them
Rights / fairness	Disparate impact, accessibility, exclusion, due-process / appeal concerns
Privacy	Minimisation, purpose, security, leakage and reuse
Safety / security	Failure modes, misuse, abuse, adversarial risks, fallback
Transparency	Notice, explanation, documentation and user understanding
Human oversight	Who reviews, overrides or escalates? With what authority?
Controls	Technical, organisational, contractual and policy mitigations
Residual risk	What remains after mitigation and who accepts it?
Monitoring	Metrics, complaints, incidents, drift and reassessment triggers

APPENDIX B: RESPONSIBLE AI RISK REGISTER

Risk ID	Risk / Harm	Affected Group	Severity	Likelihood	Control	Owner	Residual Status
R-01			—	—			Open / Mitigated
R-02			—	—			Open / Mitigated
R-03			—	—			Open / Mitigated
R-04			—	—			Open / Mitigated

APPENDIX C: MODEL / SYSTEM DOCUMENTATION CHECKLIST

- Purpose, users and prohibited uses are documented.
- Training / evaluation data provenance is recorded where available.
- Known limitations and subgroup performance are documented.
- Model / system version is traceable.
- Human oversight and fallback are defined.
- Privacy and security review is complete.
- Adversarial / misuse testing matches the use case.
- User disclosure / explanation is prepared.
- Complaint, correction and incident channels are defined.
- Vendor changes trigger reassessment.
- Monitoring metrics and retirement criteria are defined.

APPENDIX D: SAMPLE VIVA-VOCE QUESTIONS

1. What is AI ethics?
2. What is responsible AI?
3. What is the difference between an ethical principle and a legal obligation?
4. Why is fairness context-dependent?
5. What is proxy discrimination?
6. What is meaningful human oversight?
7. What is algorithmic transparency?
8. What is explainability?
9. What is an AI Impact Assessment?
10. What is TEVV?
11. What are the four core functions of the NIST AI RMF?
12. What changed in the OECD AI Principles in 2024?
13. What is the UNESCO Recommendation on the Ethics of AI?
14. What is the basic risk-based approach of the EU AI Act?
15. What are general-purpose AI models under the EU AI Act?
16. What is the role of NITI Aayog's Responsible AI approach?
17. How does the DPDP framework relate to AI?
18. Why must the phased commencement of the DPDP Rules be checked?
19. What is automation bias?
20. Why is vendor governance important for AI?
21. What risks are specific or especially relevant to generative AI?
22. What is a residual risk?
23. What was the main fairness concern in the CareerBridge case study?
24. Why was “browse all eligible internships” included as a control?
25. Why is responsible AI considered a lifecycle process?

APPENDIX E: GLOSSARY OF TERMS

Term	Meaning
AI assurance	Evidence-based activities used to build confidence that an AI system meets defined requirements.
AI Impact Assessment	Structured assessment of purpose, affected groups, risks, controls and residual impacts of an AI system.
Algorithmic bias	Systematic patterns in data, design or deployment that can produce unfair or distorted outcomes.
Automation bias	Human tendency to over-rely on automated output or recommendations.
Contestability	Ability of affected people to question, correct or appeal a consequential outcome.
Explainability	Ability to provide useful reasons or understanding of AI behaviour or decisions.
Fairness	Context-dependent principle concerned with unjustified unequal treatment or outcomes.
GPAI	General-purpose AI model capable of being used across many downstream tasks.
Human oversight	Meaningful human ability to review, intervene, override or escalate.
Model card	Structured documentation about model purpose, performance, limitations and evaluation.
Proxy variable	Feature correlated with another protected or sensitive attribute and capable of recreating its effect.
Red teaming	Adversarial testing intended to discover misuse, failure modes or security/safety weaknesses.
Residual risk	Risk remaining after controls are applied.
Responsible AI	Governance and engineering practices intended to develop and use AI in a trustworthy, rights-respecting and accountable manner.
TEVV	Testing, Evaluation, Verification and Validation.
Transparency	Disclosure and documentation that make AI use, purpose, limits and governance appropriately visible.